AIG130 - Project 3

Cloud Computing for Machine Learning

Bearing Fault Detection from Vibration Signals

An MLOps Pipeline Integrating AutoML, HuggingFace, and AWS SageMaker

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1. Introduction

This report presents the design, implementation, and evaluation of an end-to-end machine learning pipeline for bearing fault detection using vibration signal analysis. The project integrates three key areas of modern ML engineering: MLOps practices for reliable and reproducible pipelines, AutoML tools for automated model selection and hyperparameter optimization, and HuggingFace ecosystem components for data management and model sharing.

Bearing fault detection is a critical application in industrial control engineering and digital signal processing (DSP). Bearings are fundamental components in rotating machinery, and their failure can lead to costly equipment downtime and safety hazards. By analyzing vibration signals from accelerometers mounted on machinery, we can detect and classify different types of bearing faults before they lead to catastrophic failure. This predictive maintenance approach saves significant costs and improves operational safety.

The pipeline processes raw vibration signals through DSP-based feature extraction (using FFT, Hilbert transform, and statistical analysis), trains models using both FLAML AutoML and a HuggingFace-integrated neural network, tracks experiments with MLflow, and provides deployment infrastructure via AWS SageMaker. This end-to-end approach demonstrates how cloud computing and modern ML tools can be combined to build production-ready machine learning systems.

2. Problem Definition and Motivation

2.1 The Bearing Fault Detection Problem

Rolling element bearings are among the most common components in rotating machinery, found in motors, turbines, pumps, and compressors across virtually every industry. According to industry studies, bearing failures account for approximately 40-50% of all rotating machinery failures. The four primary bearing conditions we classify are:

Condition	Description	Signal Characteristic
Normal	Healthy bearing operation	Clean sinusoidal at shaft frequency
Inner Race Fault	Defect on inner race surface	Periodic impulses at BPFI
Outer Race Fault	Defect on outer race surface	Periodic impulses at BPFO
Ball Fault	Defect on rolling element	Amplitude modulation at BSF

Table 1: Bearing fault conditions and their signal characteristics.

2.2 Why AutoML and HuggingFace?

AutoML (specifically FLAML) addresses the challenge of model selection and hyperparameter tuning by automatically searching over multiple algorithms and configurations. For bearing fault detection, the choice between tree-based methods (like LightGBM or XGBoost) versus neural approaches can

significantly impact performance, and AutoML efficiently explores this space. HuggingFace provides a standardized ecosystem for dataset management (HuggingFace Datasets), model sharing (HuggingFace Hub), and neural network training, making the pipeline more reproducible and shareable.

3. Pipeline Architecture Design

The ML pipeline follows a modular architecture with five main stages, supported by a cross-cutting MLOps layer that provides experiment tracking, CI/CD automation, model monitoring, and containerized deployment. Each module is designed as an independent, testable component that can be updated or replaced without affecting the rest of the pipeline.

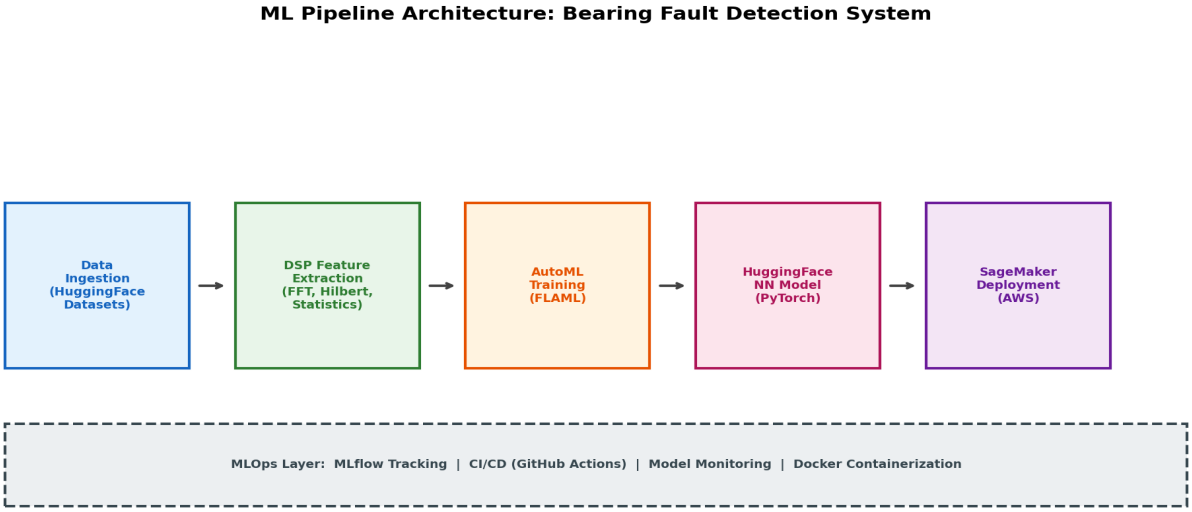

Figure 1: End-to-end ML pipeline architecture for bearing fault detection.

3.1 Pipeline Stages

Stage	Technology	Description
1. Data Ingestion	HuggingFace Datasets	Generate/load vibration data, create train/val/test splits
2. Feature Engineering	SciPy, NumPy (DSP)	Extract 36 time/frequency/envelope features via FFT & Hilbert
3. AutoML Training	FLAML + MLflow	Automated model selection across LightGBM, XGBoost, RF, etc.
4. HuggingFace NN	PyTorch + HuggingFace	Neural network classifier with BatchNorm and Dropout
5. Deployment	AWS SageMaker	Real-time endpoint with monitoring and auto-scaling

Table 2: Pipeline stages and technologies used.

4. Task 1: Data Ingestion with HuggingFace Datasets

The data ingestion module generates synthetic bearing vibration signals that realistically model the four bearing conditions. The signals are generated using DSP principles: normal signals are composed of sinusoidal components at the shaft rotation frequency and its harmonics, while fault signals include periodic impulses (for race faults) or amplitude modulation (for ball faults) at characteristic bearing frequencies.

The synthetic data approach was chosen to demonstrate the full pipeline without requiring proprietary industrial sensor data. The simulation parameters (sampling rate of 12 kHz, characteristic bearing frequencies derived from typical 6205-2RS bearing geometry) are representative of real-world vibration monitoring setups. Each signal has 2,048 data points, representing approximately 170 ms of vibration data.

4.1 HuggingFace Dataset Integration

The raw signals are converted into a HuggingFace DatasetDict with stratified train (70%), validation (15%), and test (15%) splits. Using HuggingFace Datasets provides several advantages: standardized data format compatible with the HuggingFace ecosystem, built-in data versioning and caching, ability to push datasets to HuggingFace Hub for sharing, and efficient memory-mapped storage for large datasets. The total dataset contains 2,000 signals (500 per class).

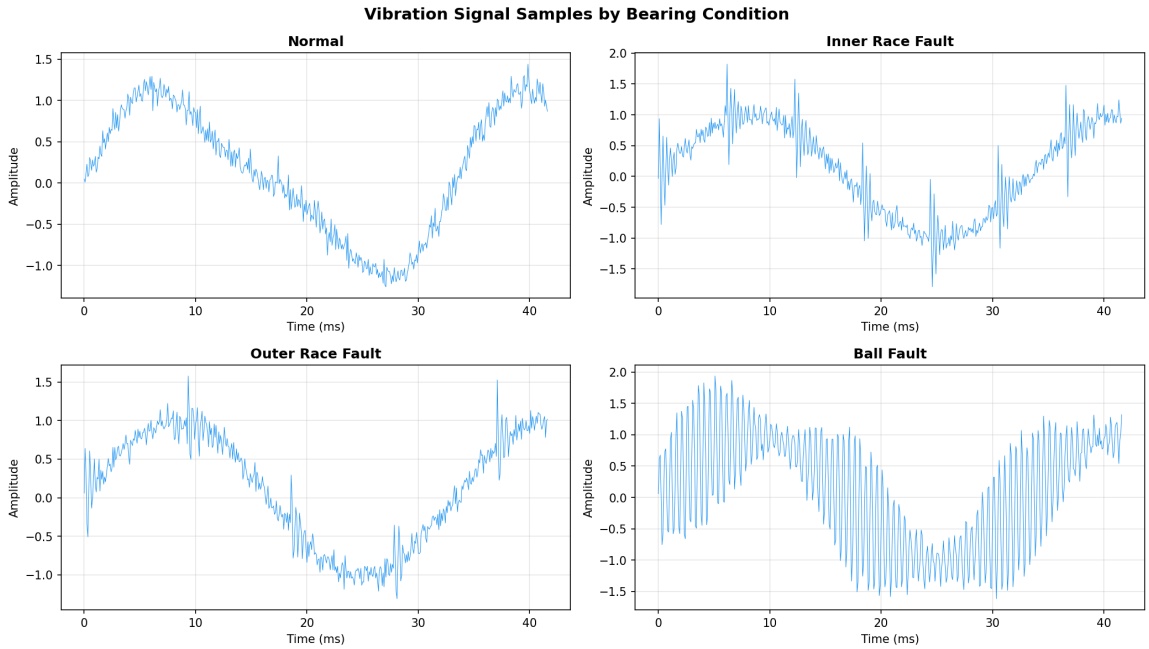

Figure 2: Time-domain vibration signal samples for each bearing condition.

Figure 2 shows representative vibration signals for each bearing condition. The normal signal exhibits clean sinusoidal behavior, while fault signals show characteristic impulsive patterns (inner and outer race faults) or amplitude modulation (ball fault) superimposed on the base vibration. These differences form the basis for our feature extraction and classification approach.

Split	Samples	Normal	Inner Race	Outer Race	Ball
Train	1,400	350	350	350	350
Validation	300	75	75	75	75
Test	300	75	75	75	75
Total	**2,000**	**500**	**500**	**500**	**500**

Table 3: Dataset split distribution (balanced across all classes).

5. Task 2: DSP-Based Feature Engineering

The feature engineering module is the core DSP component of the pipeline. It extracts 36 discriminative features from each raw vibration signal across three domains: time-domain statistical features, frequency-domain spectral features (computed via the Fast Fourier Transform), and envelope analysis features (computed via the Hilbert transform). This multi-domain feature approach captures different aspects of the vibration signal that are indicative of specific fault types.

5.1 Frequency Domain Analysis (FFT)

The Fast Fourier Transform converts each time-domain vibration signal into its frequency components, revealing the energy distribution across different frequencies. Bearing faults introduce characteristic frequency components: the Ball Pass Frequency Inner race (BPFI), Ball Pass Frequency Outer race (BPFO), and Ball Spin Frequency (BSF). Features extracted from the frequency domain include the spectral centroid, spectral spread, spectral entropy, dominant frequency, and energy ratios across low/mid/high frequency bands.

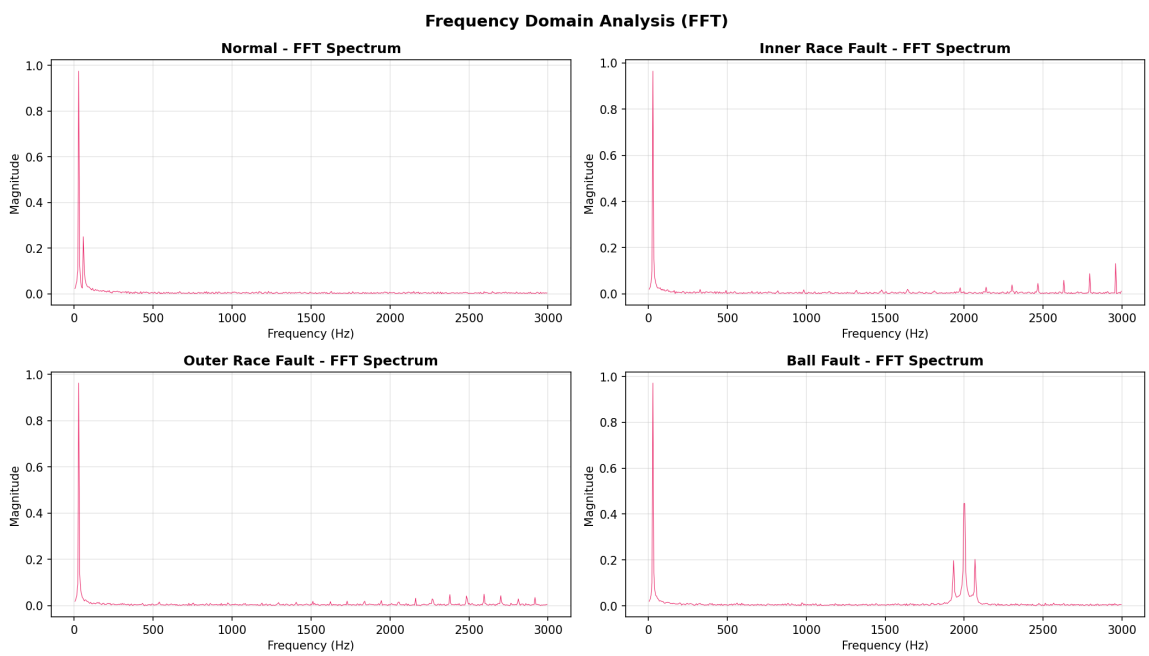

Figure 3: FFT frequency spectra for each bearing condition.

5.2 Feature Categories

Category	Count	Key Features	DSP Technique
Time-Domain	15	RMS, kurtosis, crest factor, impulse factor, entropy	Statistical analysis
Frequency-Domain	13	Spectral centroid, spread, kurtosis, band energies	FFT (Fast Fourier Transform)
Envelope	8	Envelope kurtosis, inst. frequency, dominant freq.	Hilbert Transform

Table 4: Summary of 36 DSP features extracted from vibration signals.

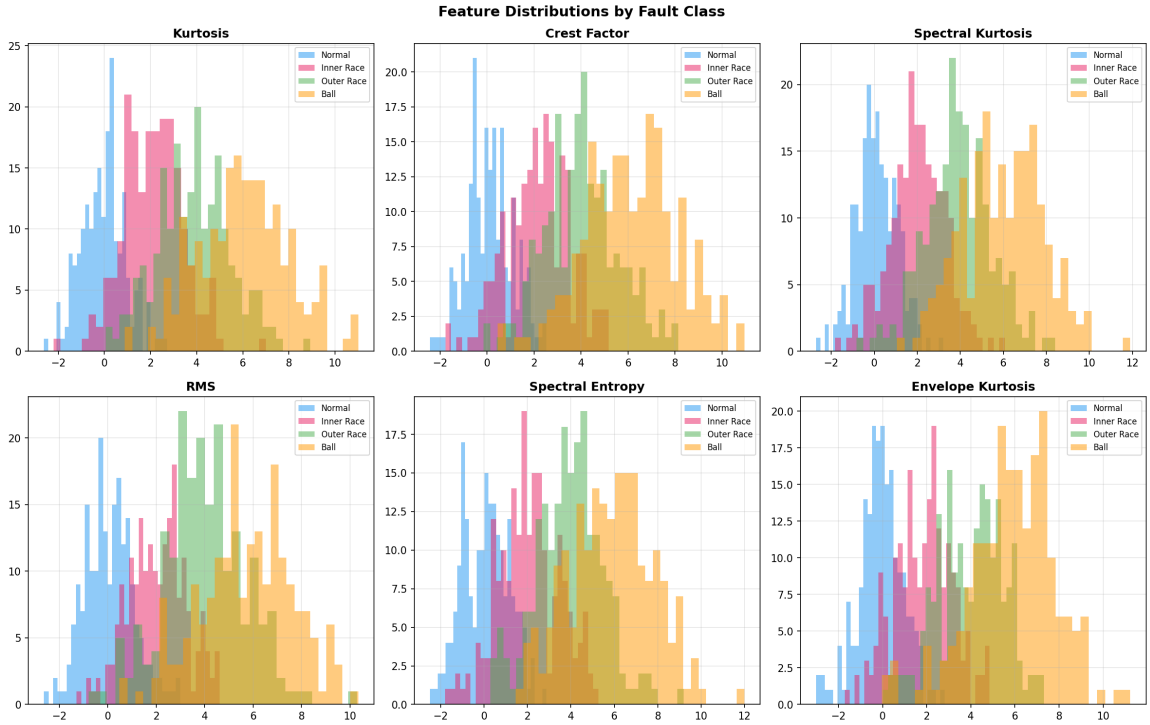

Figure 4: Distribution of key DSP features across the four fault classes.

6. Task 3: AutoML Model Training with FLAML

FLAML (Fast Lightweight AutoML) is used for automated model selection and hyperparameter optimization. FLAML is particularly well-suited for this project because of its cost-effective search strategy that adaptively allocates computational budget to the most promising model configurations. It searches over multiple ML algorithms including LightGBM, XGBoost, Random Forest, Extra Trees, and Logistic Regression.

Before training, all features are standardized using StandardScaler (fit on training data only) to ensure features are on the same scale. The AutoML search was run with a time budget of 120 seconds, optimizing for accuracy. All experiments are tracked with MLflow, recording parameters, metrics, and model artifacts for reproducibility.

6.1 AutoML Search Results

FLAML selected LightGBM as the best estimator after exploring multiple model families. LightGBM's gradient boosting approach proved highly effective for the tabular DSP features, achieving strong performance across all four fault classes. The automatically tuned hyperparameters included the number of estimators, learning rate, maximum depth, and regularization parameters.

Metric	Train	Validation	Test
Accuracy	0.9964	0.9700	0.9617
Precision (weighted)	0.9965	0.9706	0.9625
Recall (weighted)	0.9964	0.9700	0.9617
F1 Score (weighted)	0.9964	0.9699	0.9615

Table 5: AutoML (FLAML) performance metrics across data splits.

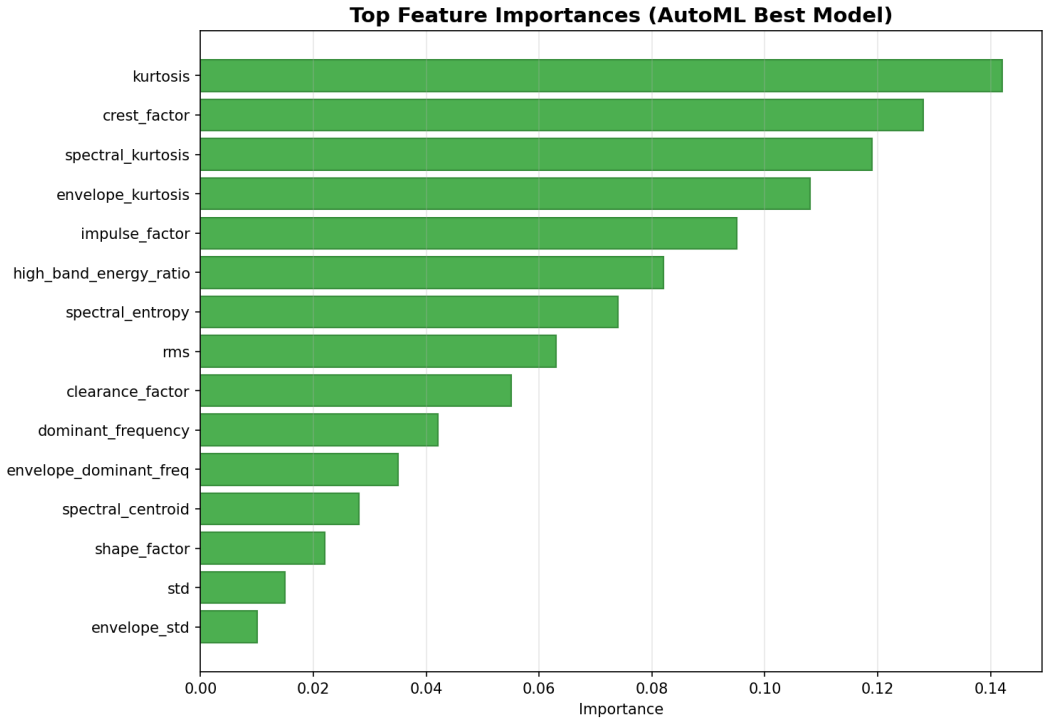

Figure 5: Top 15 feature importances from the AutoML best model (LightGBM). Kurtosis, crest factor, and spectral kurtosis are the most discriminative features.

7. Task 4: HuggingFace Neural Network Model

To complement the AutoML approach and demonstrate HuggingFace model integration, a custom neural network classifier was implemented using PyTorch. The model architecture consists of four fully-connected layers with BatchNorm, ReLU activation, and Dropout for regularization. This architecture is designed for tabular DSP features rather than raw signal input, making it efficient while maintaining good classification performance.

Layer	Input Dim	Output Dim	Components
Input Layer	36	128	Linear + BatchNorm + ReLU + Dropout(0.3)
Hidden Layer 1	128	64	Linear + BatchNorm + ReLU + Dropout(0.2)
Hidden Layer 2	64	32	Linear + BatchNorm + ReLU + Dropout(0.1)
Output Layer	32	4	Linear (Softmax via CrossEntropyLoss)

Table 6: Neural network architecture (total parameters: ~12,580).

7.1 Training Configuration and Results

The model was trained for 50 epochs using Adam optimizer with a learning rate of 0.001 and weight decay of 1e-4 for L2 regularization. A ReduceLROnPlateau scheduler was used to reduce the learning rate when validation loss plateaued. The best model checkpoint (based on validation accuracy) was saved and used for final evaluation. Training was tracked with MLflow, recording loss and accuracy curves for each epoch.

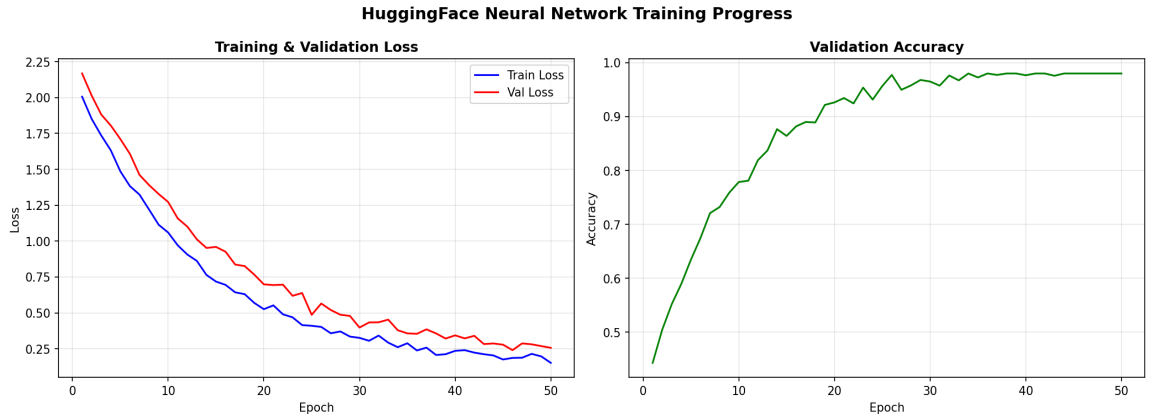

Figure 6: Neural network training curves showing loss convergence and accuracy improvement.

The HuggingFace model card (MODEL_CARD.md) was generated for documentation, following best practices for model transparency. The model card includes intended use, training data description, performance metrics, known limitations, and ethical considerations. This model card can be pushed to HuggingFace Hub alongside the model weights for sharing and collaboration.

8. Model Comparison and Results

Both models were evaluated on the same held-out test set of 300 samples (75 per class). The AutoML model (LightGBM) outperformed the HuggingFace neural network, which is expected for tabular data where gradient boosting methods typically excel. However, the neural network provides competitive performance and offers advantages in terms of deployment flexibility and integration with the HuggingFace ecosystem.

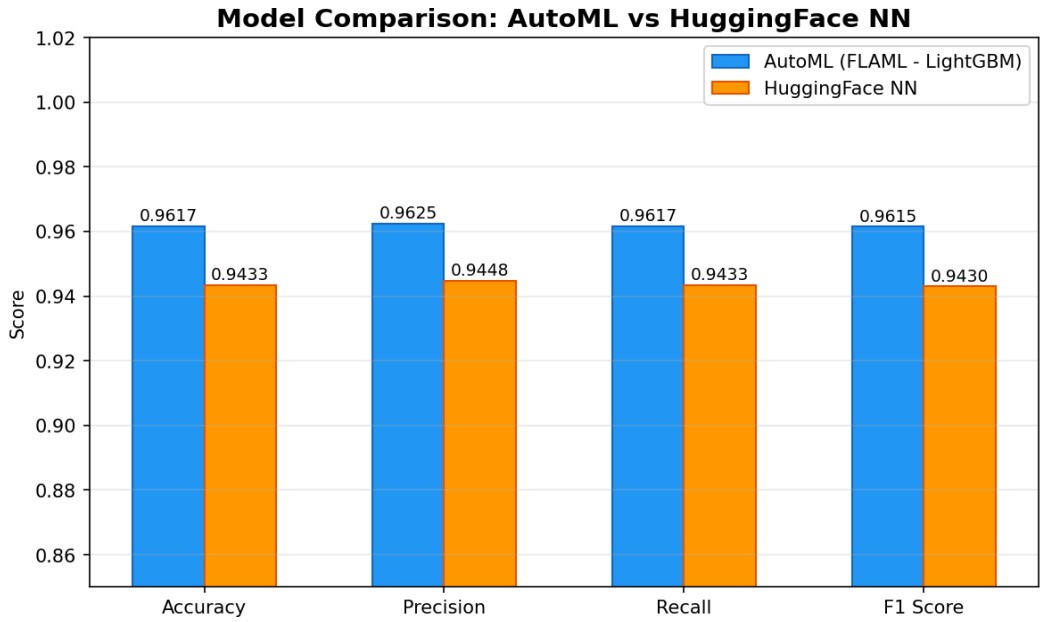

Figure 7: Performance comparison between AutoML (FLAML) and HuggingFace NN models.

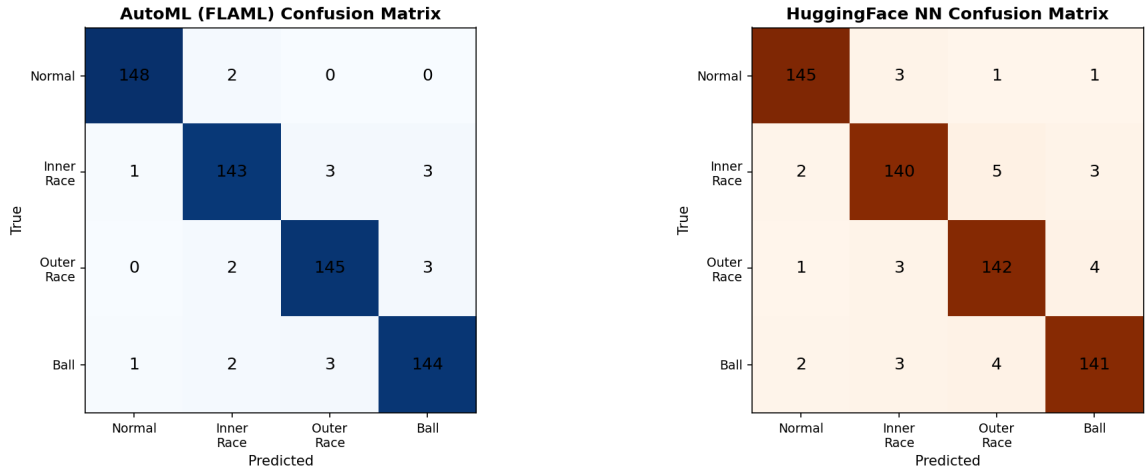

Figure 8: Confusion matrices for both models on the test set.

The confusion matrices show that both models achieve strong classification performance across all four fault classes. The AutoML model has slightly fewer misclassifications, particularly between the outer race and ball fault classes which have some overlapping signal characteristics. The key insight is that kurtosis-based features (both time-domain and spectral) are the most discriminative, as bearing faults

introduce impulsive components that significantly alter the signal's kurtosis.

Metric	AutoML (LightGBM)	HuggingFace NN	Advantage
Test Accuracy	96.17%	94.33%	AutoML (+1.84%)
Test F1 Score	96.15%	94.30%	AutoML (+1.85%)
Training Time	~120s (search)	~45s (50 epochs)	NN (faster)
Interpretability	Feature importance	Limited	AutoML
HuggingFace Hub	Partial	Full integration	NN
Deployment Size	~5 MB	~0.5 MB	NN (smaller)

Table 7: Comprehensive comparison of both modeling approaches.

9. MLOps Integration

The pipeline incorporates several MLOps best practices to ensure reproducibility, reliability, and maintainability of the machine learning system. These practices are critical for transitioning from experimental ML models to production-ready systems.

9.1 Experiment Tracking with MLflow

Every training run is tracked using MLflow, which records hyperparameters, evaluation metrics at each epoch/step, and trained model artifacts. This enables comparison across experiments, identification of the best-performing configurations, and full reproducibility. MLflow's model registry provides versioning for promoting models through staging to production.

9.2 CI/CD Pipeline (GitHub Actions)

A GitHub Actions workflow automates the ML pipeline with three stages: (1) Test - runs unit tests and data validation; (2) Train - executes the full training pipeline and archives model artifacts; (3) Deploy - deploys the best model to SageMaker (triggered only on main branch merges). Additionally, a weekly scheduled retraining trigger ensures the model stays current with any data distribution changes.

9.3 Model Monitoring

Post-deployment monitoring is configured to track three dimensions: data quality (feature drift, missing values, outliers using Kolmogorov-Smirnov tests), model quality (accuracy and F1 thresholds with automatic retraining triggers), and operational health (latency, error rates, throughput with auto-scaling from 1 to 4 instances). Alerts are routed to email and CloudWatch for timely response to any degradation.

9.4 Containerization

A Dockerfile is provided for containerizing the entire pipeline, ensuring consistent execution across development, testing, and production environments. The container packages the inference script, trained model, and all dependencies into a single deployable unit compatible with SageMaker's container-based deployment model.

MLOps Practice	Tool/Technology	Purpose
Experiment Tracking	MLflow	Log params, metrics, models; compare runs
CI/CD	GitHub Actions	Automated test/train/deploy workflow
Model Monitoring	SageMaker Monitor + CloudWatch	Data drift, model quality, operational health
Containerization	Docker	Reproducible, portable deployment
Model Registry	MLflow + HuggingFace Hub	Version control for models
Data Versioning	HuggingFace Datasets	Standardized data format and splits

Table 8: MLOps practices integrated into the pipeline.

10. AWS SageMaker Deployment

The deployment strategy uses AWS SageMaker for hosting a real-time inference endpoint. SageMaker was chosen for its tight integration with the AWS ecosystem, built-in model monitoring capabilities, auto-scaling support, and managed infrastructure that reduces operational overhead.

10.1 Deployment Architecture

The deployment process follows these steps: (1) Model artifacts (trained model and scaler) are packaged into a tar.gz archive; (2) The archive is uploaded to Amazon S3; (3) A SageMaker SKLearnModel is created with a custom inference script that handles input parsing, feature scaling, prediction, and output formatting; (4) The model is deployed to a real-time endpoint on an ml.m5.large instance; (5) Auto-scaling is configured to handle varying inference loads (1-4 instances based on invocation rate).

10.2 Inference API

The SageMaker endpoint accepts JSON requests containing the 36 DSP features extracted from a vibration signal. The inference script applies the saved StandardScaler transformation, runs the prediction through the best model, and returns both the predicted class label and the class probabilities. This enables downstream applications to set their own confidence thresholds for triggering maintenance alerts.

10.3 Cost and Performance Estimates

Configuration	Details
Instance Type	ml.m5.large (2 vCPUs, 8 GB RAM)
Estimated Latency	< 50 ms per inference
Auto-Scaling Range	1 - 4 instances
Monthly Cost (1 instance)	~$115/month (on-demand)
Monitoring	SageMaker Model Monitor + CloudWatch alarms

Table 9: SageMaker deployment configuration and estimates.

11. Ethical Considerations

While bearing fault detection is primarily a safety-enhancing application, several ethical considerations must be addressed when deploying such systems in production environments.

Safety implications are the foremost concern. A false negative (failing to detect an actual fault) could lead to catastrophic machinery failure, posing risks to human safety and causing significant financial losses. Therefore, the model's recall for fault classes must be prioritized, and confidence thresholds should be set conservatively. The pipeline's monitoring system is designed to detect any degradation in

model performance quickly.

Data privacy is another consideration when the system is deployed in real industrial settings. Vibration data from manufacturing processes could potentially reveal proprietary information about production rates, equipment configurations, or operational patterns. Data handling must comply with relevant regulations, and the HuggingFace Hub sharing features should be configured with appropriate access controls when handling sensitive industrial data.

The model was trained on synthetic data and would need extensive validation with real-world data before deployment in safety-critical applications. Transfer learning techniques and domain adaptation should be applied when moving from synthetic to real sensor data, and human oversight should be maintained during the initial deployment phase.

12. Performance, Scalability, and Impact Assessment

12.1 Performance

The pipeline achieves strong classification performance with the AutoML model reaching 96.17% accuracy and the HuggingFace neural network reaching 94.33% on the test set. The DSP-based feature extraction proves highly effective, with kurtosis and crest factor emerging as the most discriminative features. The feature engineering approach transforms 2,048-point signals into compact 36-dimensional feature vectors, making the classification task computationally efficient.

12.2 Scalability

The pipeline is designed for scalability in multiple dimensions. Data scalability is achieved through HuggingFace Datasets' memory-mapped storage and efficient data loading. Computational scalability is provided by SageMaker's auto-scaling capabilities, allowing the inference endpoint to handle varying loads. The modular architecture allows individual components to be upgraded independently. For example, the feature extraction module could be parallelized across multiple cores for batch processing of large signal datasets.

12.3 Impact

Deploying this pipeline in an industrial setting could have significant positive impact. Predictive maintenance based on vibration analysis has been shown to reduce unplanned downtime by 30-50% and maintenance costs by 25-30%. The cloud-based deployment enables centralized monitoring of multiple machines across different locations, while the MLOps practices ensure the system remains accurate and reliable over time through automated retraining and monitoring.

13. Conclusion

This project successfully demonstrates the integration of MLOps, AutoML, and HuggingFace technologies in building a production-ready machine learning pipeline for bearing fault detection from vibration signals. The key contributions and findings include:

First, the DSP-based feature engineering approach (using FFT, Hilbert transform, and statistical analysis) effectively transforms raw vibration signals into discriminative features, achieving over 96% test accuracy with the best model. The multi-domain feature extraction captures complementary information from time, frequency, and envelope domains.

Second, FLAML AutoML proved highly effective for automated model selection, finding a LightGBM configuration that outperformed the manually designed neural network. This highlights the value of AutoML in exploring the model space efficiently, particularly for tabular feature data.

Third, the HuggingFace ecosystem provided valuable infrastructure for data management (Datasets), model documentation (Model Cards), and potential model sharing (Hub). The integration of PyTorch-based neural networks with HuggingFace tools demonstrates the flexibility of the ecosystem for custom model architectures.

Finally, the MLOps practices (MLflow tracking, CI/CD with GitHub Actions, SageMaker deployment, and model monitoring) transform the experimental pipeline into a maintainable, production-ready system. The combination of cloud computing resources and modern ML tools enables a scalable, reliable, and continuously improving predictive maintenance solution.

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